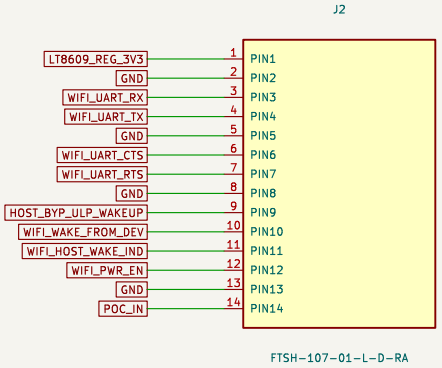
PCB-BASED RECONSTRUCTION — electrical intent is not certified

20 source footprints; module REF**0 is renamed U2 ONLY in this reconstruction.

Every named PCB net is retained literally. Net_0 is a real named net; it is not an unassigned pad.

HOST INTERFACE
J2 has 14 pads, including four GND pads.
Signal names are copied from PCB.
UART naming is not a verified host direction.
No USB/SPI/SDIO connector is present.



POWER PATH
J2.1 LT8609_REG_3V3 -> U1 ideal-diode switch -> L11 -> Untitled rail.
There is NO LT8609 regulator on this board; the net name refers to the host.
J2.12 WIFI_PWR_EN controls U1.3; R3 provides a 100k pull-down.

KNOWN LIMITS
The original schematic is empty. Module and G1 pin functions remain PADn
until exact manufacturer documents can be checked.
Open pads are intentionally left open in this reconstruction;
ERC must expose them. No blanket NC markers or rule suppression are used.





Power flags indicate an assumed external source at J2, not a measured voltage.

02_Power
File: 02_Power.kicad_sch

03_Module_Connected
File: 03_Module_Connected.kicad_sch

04_Module_Open_Pads
File: 04_Module_Open_Pads.kicad_sch

05_Clock_Control
File: 05_Clock_Control.kicad_sch

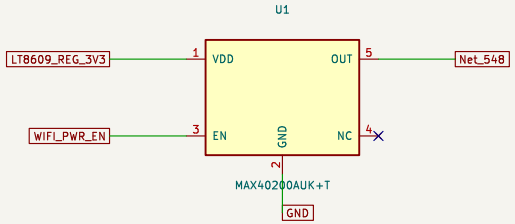
Source PCB preserved; see engineering report

Sheet: /
File: Recovered_Wifi.kicad_sch

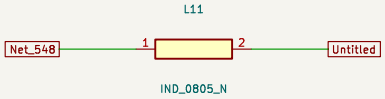
Title: Recovered_Wifi — PCB-derived review

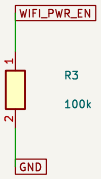
Size: A4	Date: 2026-10-23	**Rev: R1**
KiCad E.D.A. 9.0.2		Id: 1/5

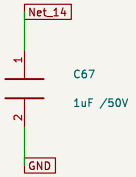
POWER / ENABLE — exact PCB pad connectivity



U1 pin map: installed KiCad MAX40200AUK symbol.
Ideal diode / enable switch, not a voltage regulator.
L11 value is only IND_0805_N: inductance, DCR,
saturation current and whether it is a bead are UNKNOWN.

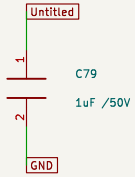


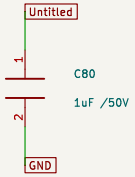


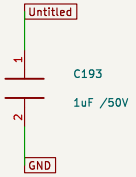


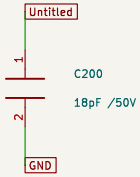
C67 connects Net_14 to GND; module pads 36 and 54 share Net_14.
Do not assume this rail is 3.3 V or an output until module pinout is verified.

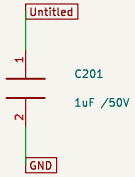
All six capacitors below share PCB net Untitled, not six independent supplies.

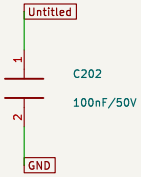












Input bulk capacitor, input TVS/fuse and host source impedance are not present as separate components.

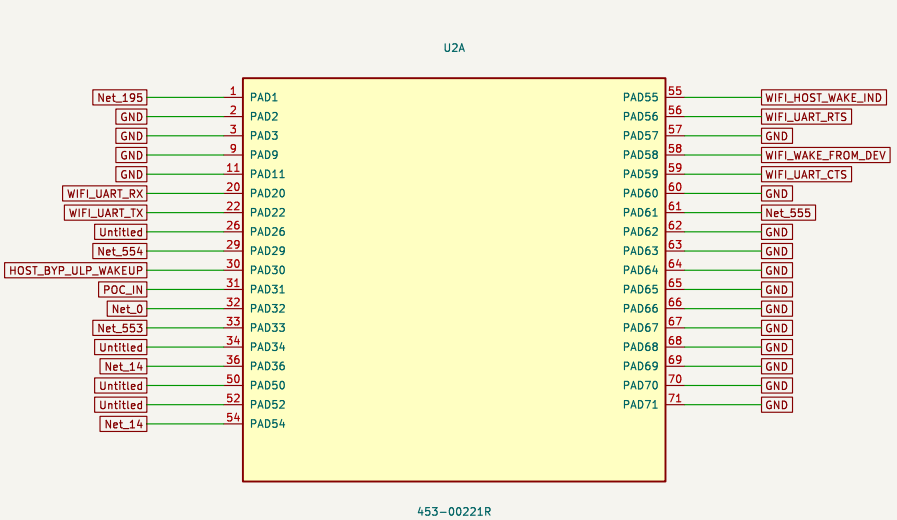
Source PCB preserved; see engineering report

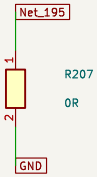
Sheet: /02_Power/
File: 02_Power.kicad_sch

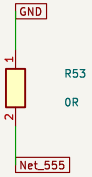
Title: 02_Power — PCB-derived review

Size: A4	Date: 2026-10-23	**Rev: R1**
KiCad E.D.A. 9.0.2		Id: 2/5

U2 — PCB-CONNECTED PADS (source REF**0 / value 453-00221R)







PADn names are deliberate: manufacturer pin functions are not verified.
All connected pin numbers and net names are preserved from source PCB.
Pin electrical types remain passive where unknown; ERC cannot certify
voltage compatibility, boot straps or UART direction for these pins.

Pad 1 -> R207 (0R) -> GND
Pad 61 -> R53 (0R) -> GND
Their functions MUST be checked before calling these RF ports or straps.

See unit B on next sheet for all physically open pads.
Ground pads 62..71 are real numbered pads, not omitted hidden power pins.

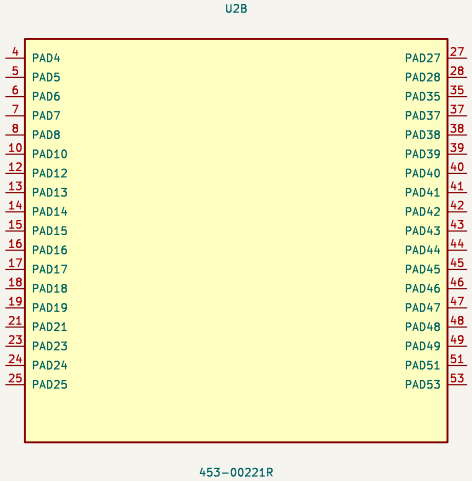
Source PCB preserved; see engineering report

Sheet: /03_Module_Connected/
File: 03_Module_Connected.kicad_sch

Title: 03_Module_Connected — PCB-derived review

Size: A4	Date: 2026-10-23	**Rev: R1**
KiCad E.D.A. 9.0.2		Id: 3/5

U2 — PCB-OPEN PADS, NOT VERIFIED MANUFACTURER NC



These pads have no assigned PCB net and no observed track/pad contact.
Open schematic pins reproduce that observation; they are NOT marked NC.
Their necessity for boot, power, RF or programming is still unresolved.
ERC pin-not-connected errors are retained intentionally for review.

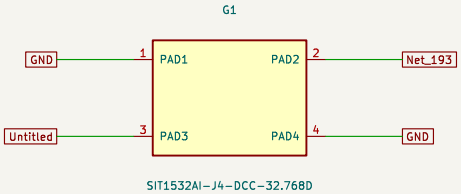
Source PCB preserved; see engineering report

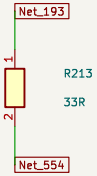
Sheet: /04_Module_Open_Pads/
File: 04_Module_Open_Pads.kicad_sch

Title: 04_Module_Open_Pads — PCB-derived review

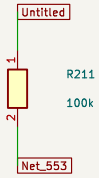
Size: A4	Date: 2026-10-23	**Rev: R1**
KiCad E.D.A. 9.0.2		Id: 4/5

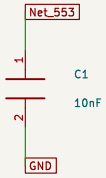
CLOCK / CONTROL / POC — topology preserved, functional names conditional

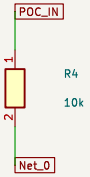


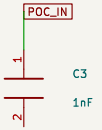


G1 has value SiT1532AI-J4-DCC-32.768D.
PCB pad 2 -> Net_193 -> R213 33R -> Net_554 -> U2.29.
PCB pad 3 -> Untitled; pads 1 and 4 -> GND.
This is observed wiring, not a verified oscillator pinout.









POC_IN -> U2.31 and C3.1.
R4 10k links POC_IN to Net_0 -> U2.32.
C3.2 is physically open in source PCB: no return path.
Do NOT silently connect it to GND in reconstruction.

R211 100k + C1 10nF: nominal RC = 1 ms.
Node Net_553 goes to U2.33.
Reset/strap function remains conditional;
threshold and internal pulls are not verified.

Source PCB preserved; see engineering report

Sheet: /05_Clock_Control/
File: 05_Clock_Control.kicad_sch

Title: 05_Clock_Control — PCB-derived review

Size: A4	Date: 2026-10-23	**Rev: R1**
KiCad E.D.A. 9.0.2		Id: 5/5